THERMAL PROPERTIES OF MATTER

NCERT Class 11 Physics — Formula, Derivations & Solving Methods

Exam Revision Sheet
Target: CBSE / JEE Main / NEET
Theme: Scientific Navy & Teal

1. TEMPERATURE SCALES & GAS LAWS

CORE BASICS

Temperature Scale Conversion Relation

EXAM FAVOURITE

$$\frac{t_F - 32}{180} = \frac{t_C}{100} = \frac{T - 273.15}{100}$$

Symbols:
t_C: Temperature in Celsius ($^\circ$C)
t_F: Temperature in Fahrenheit ($^\circ$F)
T: Absolute Temperature in Kelvin (K)

Key Constants & Relations:
1 Division on t_C scale = 1.8 Divisions on t_F scale
Absolute Zero: $0\ \text{K} = -273.15^\circ\text{C} = -459.67^\circ\text{F}$
Triple Point of Water: $T_{TP} = 273.16\ \text{K} = 0.01^\circ\text{C}$

Ideal Gas Equation & Vol. Expansivity

$$PV = \mu RT \implies \alpha_V = \frac{1}{T}$$

Derivation for Ideal Gas α_V:
At constant pressure P: $P\Delta V = \mu R\Delta T$
Divide by $PV = \mu RT \implies \frac{\Delta V}{V} = \frac{\Delta T}{T}$

$$\alpha_V = \frac{1}{V}\frac{\Delta V}{\Delta T} = \frac{1}{T} \quad (\text{At } 0^\circ\text{C}, \alpha_V \approx 3.67 \times 10^{-3}\ \text{K}^{-1})$$

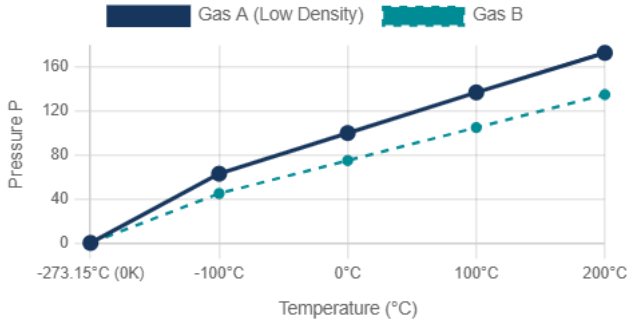


Fig 1.1: P vs T Linear Extrapolation to Absolute Zero

2. THERMAL EXPANSION IN SOLIDS & LIQUIDS

MUST KNOW

Coefficients of Expansion ($\alpha_l, \alpha_a, \alpha_v$)

$$\Delta l = \alpha_l l \Delta T, \quad \Delta A = \alpha_a A \Delta T, \quad \Delta V = \alpha_v V \Delta T$$

$$\alpha_l : \alpha_a : \alpha_v = 1 : 2 : 3 \implies \alpha_a = 2\alpha_l, \quad \alpha_v = 3\alpha_l$$

SI Units: K^{-1} or $(^\circ\text{C})^{-1}$

Thermal Stress in Fixed Rod:

$$\text{Stress} = \frac{F}{A} = Y\left(\frac{\Delta l}{l}\right) = Y\alpha_l \Delta T$$

$$\text{Compressive Force: } F = AY\alpha_l \Delta T$$

Density Variation with Temp:

$$\rho' = \frac{\rho}{1 + \alpha_v \Delta T} \approx \rho(1 - \alpha_v \Delta T)$$

(Valid for $\alpha_v \Delta T \ll 1$)

Proof / Derivation: Relationship $\alpha_v = 3\alpha_l$ for Isotropic Solids

1. Consider a cube of initial length l. Expanded length $l' = l + \Delta l = l(1 + \alpha_l \Delta T)$.
2. New Volume $V' = (l')^3 = l^3(1 + \alpha_l \Delta T)^3 = V(1 + 3\alpha_l \Delta T + 3\alpha_l^2 \Delta T^2 + \alpha_l^3 \Delta T^3)$.
3. Since $\alpha_l \sim 10^{-5}\ \mathrm{K}^{-1}$, terms containing α_l^2 and α_l^3 are negligible.
4. $\Delta V = V' - V \approx 3\alpha_l V \Delta T$. Comparing with $\Delta V = \alpha_v V \Delta T \implies \boldsymbol{\alpha_v = 3\alpha_l}$.

⚠ CAUTION: Anomalous Expansion of Water

Between $0^\circ\mathrm{C}$ and $4^\circ\mathrm{C}$, water **contracts** on heating (α_v is negative). Water has **maximum density** at $4^\circ\mathrm{C}$ ($\rho_{\max} = 1000\ \mathrm{kg/m^3}$). This allows aquatic life to survive at the bottom of frozen lakes.

3. CALORIMETRY & CHANGE OF STATE

MUST KNOW

Heat Capacity, Specific Heat & Latent Heat

$$\Delta Q = ms\Delta T, \quad \Delta Q = \mu C \Delta T, \quad \Delta Q = mL$$

Specific Heat Capacity (s):

$$s = \frac{1}{m}\frac{\Delta Q}{\Delta T} \quad [\text{SI: J kg}^{-1}\text{K}^{-1}]$$

$s_{\text{water}} = 4186 \text{ J kg}^{-1}\text{K}^{-1} \approx 1 \text{ cal g}^{-1}(^\circ\text{C})^{-1}$

Latent Heat (L):

$$L = \frac{Q}{m} \quad [\text{SI: J kg}^{-1}]$$

L_f (Ice) $= 3.33 \times 10^5$ J/kg (80 cal/g)
L_v (Steam) $= 22.6 \times 10^5$ J/kg (540 cal/g)

Numerical Solving Workflow: Principle of Calorimetry

Heat Lost by Hot Bodies = Heat Gained by Cold Bodies

Identify Phases → Check Temp Bounds → Include mL if Phase Changes → Equate $Q_{\text{lost}} = Q_{\text{gained}}$

Note: Always verify if available heat from hot body is sufficient to melt/vaporize the cold body completely.

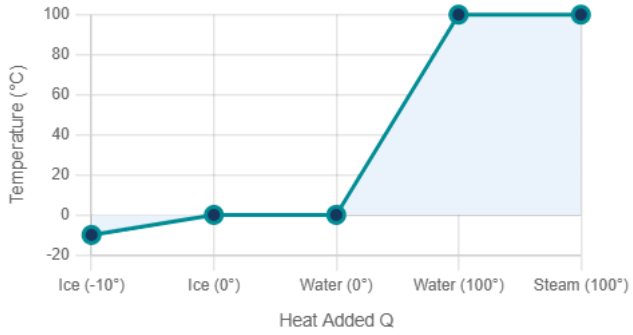


Fig 3.1: Heating Curve of Water (T vs Q)

4. HEAT TRANSFER: CONDUCTION, CONVECTION & RADIATION

EXAM FAVOURITE

Thermal Conduction (Fourier's Law) & Thermal Resistance

$$H = \frac{dQ}{dt} = KA\frac{T_C - T_D}{L} = \frac{\Delta T}{R_{th}} \quad \text{where} \quad R_{th} = \frac{L}{KA}$$

Combination Type	Equivalent Thermal Resistance ($R_{th,eq}$)	Equivalent Conductivity (K_{eq})
Series Combination ($L = L_1 + L_2, A_1 = A_2$)	$R_{eq} = R_1 + R_2 = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$	$K_{eq} = \frac{L_1+L_2}{\frac{L_1}{K_1}+\frac{L_2}{K_2}}$ $\left[\text{If } L_1 = L_2 : K_{eq} = \frac{2K_1K_2}{K_1+K_2}\right]$
Parallel Combination ($A = A_1 + A_2, L_1 = L_2$)	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$	$K_{eq} = \frac{K_1A_1+K_2A_2}{A_1+A_2}$ $\left[\text{If } A_1 = A_2 : K_{eq} = \frac{K_1+K_2}{2}\right]$

Radiation Laws: Stefan-Boltzmann & Wien's Displacement Law

$$H = e\sigma AT^4, \quad H_{\text{net}} = e\sigma A(T^4 - T_s^4), \quad \lambda_m T = b$$

Stefan-Boltzmann Law:
$\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\text{K}^{-4}$ (Stefan's Constant)
$e = \text{Emissivity}\ (0 \le e \le 1); \quad e = 1$ for Blackbody

Wien's Displacement Law:
$b = 2.9 \times 10^{-3}\ \text{m K}$ (Wien's Constant)
$\lambda_m \propto \frac{1}{T} \implies$ As $T \uparrow$, Peak λ_m shifts to shorter λ.

5. NEWTON'S LAW OF COOLING

SHORTCUT METHODS

Differential & Approximate Forms

$$\text{Exact: } \frac{dT}{dt} = -k(T - T_s) \implies \ln\left(\frac{T_2 - T_s}{T_1 - T_s}\right) = -kt$$

$$\text{Approximate (For Numericals): } \frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_s\right)$$

Conditions for Validity:
1. Temperature difference $(T - T_s)$ must be small ($\lesssim 30^\circ\text{C}$).
2. Heat loss occurs predominantly via radiation & forced convection.

Linear Graph Verification:
Plot of $\ln(T - T_s)$ vs time t gives a **straight line** with slope $= -k$.

Step-by-Step Solved Exam Numerical (Newton's Law of Cooling)

Problem: A body cools from 80°C to 50°C in 5 minutes in a room at 20°C. Calculate the time taken to cool from 60°C to 30°C.

1. **Case 1:** $T_1 = 80^\circ\text{C}, T_2 = 50^\circ\text{C}, T_s = 20^\circ\text{C}, t = 5\ \text{min}$.
   Average $T_{\text{avg}} = \frac{80+50}{2} = 65^\circ\text{C}$.
   Formula: $\frac{80-50}{5} = K(65 - 20) \implies \frac{30}{5} = K(45) \implies K = \frac{6}{45} = \frac{2}{9}\ \text{min}^{-1}$.
2. **Case 2:** $T_1 = 60^\circ\text{C}, T_2 = 30^\circ\text{C}, T_s = 20^\circ\text{C}$.
   Average $T'_{\text{avg}} = \frac{60+30}{2} = 45^\circ\text{C}$.
   Formula: $\frac{60-30}{t'} = K(45 - 20) \implies \frac{30}{t'} = \left(\frac{2}{9}\right)(25)$.
3. **Solve for t':** $t' = \frac{30 \times 9}{50} = \mathbf{5.4\ minutes = 5\ min\ 24\ s}$.

6. PHYSICAL CONSTANTS & DIMENSIONAL FORMULAS

MUST KNOW

Physical Quantity	Symbol	SI Unit	Dimensional Formula
Thermal Conductivity	K	$\text{W m}^{-1}\text{K}^{-1}$ or $\text{J s}^{-1}\text{m}^{-1}\text{K}^{-1}$	$[\text{M L T}^{-3}\text{K}^{-1}]$
Specific Heat Capacity	s	$\text{J kg}^{-1}\text{K}^{-1}$	$[\text{L}^2\text{T}^{-2}\text{K}^{-1}]$
Molar Specific Heat	C	$\text{J mol}^{-1}\text{K}^{-1}$	$[\text{M L}^2\text{T}^{-2}\text{K}^{-1}\text{mol}^{-1}]$
Latent Heat	L	J kg^{-1}	$[\text{L}^2\text{T}^{-2}]$
Stefan-Boltzmann Constant	σ	$\text{W m}^{-2}\text{K}^{-4}$	$[\text{M T}^{-3}\text{K}^{-4}]$
Wien's Constant	b	m K	[L K]